Ash Yuxiang Chen

Machine Learning Engineer / Computer Vision + MLOps

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EDUCATION

University of Waterloo **2022 – 2027**

BAsc. Honours Systems Design Engineering, Artificial Intelligence Option cGPA: 3.8/4.0

Minor: Philosophy, AI Ethics Specialization

Coursework: Deep Learning, Computer Vision, Autonomous Systems

SKILLS

ML/AI: PyTorch, Keras, TensorFlow, ONNX, Scikit-learn, NumPy, Pandas, OpenCV, Hugging Face

MLOps: Model Deployment, ONNX Conversion, Edge Inference (Jetson Orin), TensorRT, Docker, MLFlow

Languages: Python, C/C++, JavaScript/TypeScript, MATLAB, SQL

Cloud & Tools: GCP, AWS, Azure, Git, Linux/Bash, REST APIs, MongoDB, Firebase

WORK EXPERIENCE

Machine Learning Engineer - IBM, Venturelab Toronto, ON | **May. 2025 – Aug. 2025**

- Improved segmentation model F1 from **0.52 to 0.73 (+21 points)**, as measured by held-out test set, by fine-tuning EfficientNet-B3 encoder with Dice+BCE composite loss in TensorFlow and stratified k-fold cross-validation
- Reduced model debugging cycles from 5 days to 2 days per iteration by building evaluation pipelines with Scikit-learn metrics (F1, IoU, precision-recall curves) and failure case visualization
- Achieved 24 FPS inference on edge hardware (up from 8 FPS baseline) by converting **Keras models to ONNX** and optimizing with TensorRT for NVIDIA Jetson Orin deployment
- Enabled exact experiment reproduction across team by implementing **MLflow tracking** with fixed random seeds, versioned model checkpoints, and standardized TensorFlow data loaders

Undergrad Research Assistant - University of Waterloo Waterloo, ON | **Jan. 2024 – Apr 2024**

- Contributed to grant proposal supporting **$121,220 NRC-funded research** on deep reinforcement learning verification for autonomous vehicle safety using PyTorch, Gymnasium, and CARLA simulator
- Achieved 95% policy coverage across 50+ adversarial scenarios, by implementing PPO and SAC algorithms with custom reward shaping in PyTorch
- Reduced experiment runtime from 12 hours to 4 hours per evaluation cycle by engineering parallelized data pipelines using NumPy, Pandas, and Python multiprocessing

Firmware Engineer - Motorola Solutions Toronto, ON | **Sep. 2024 – Dec. 2024**

- Improved audio classification F1 from 0.78 to 0.90 (+12 points) by implementing **real-time feature extraction (FFT, MFCCs)** and inference optimization in C and MATLAB
- Reduced inference latency from 45ms to 32ms per sample end-to-end processing time, by profiling and optimizing memory-efficient signal processing pipelines for embedded deployment

Simulation Engineer - Magna International – OtO Inc. Vaughan, ON | **Sep. 2022 – Dec. 2022**

- Deployed **CNN-based weather detection to 200 beta users**, by containerizing PyTorch inference pipeline on Google Cloud Platform with **Docker and automated CI/CD**
- Reduced path planning inference from 150ms to 100ms per query, as measured by p95 latency, by optimizing A* graph search with learned heuristics and NumPy vectorization

PROJECTS

SafePulse, a health and safety app (1st Place Winner)

Courier, Geotab, Firebase, Google Cloud Platform, REST APIs

- Won **1st place** among 200+ participants with emergency responder app by building scalable real-time data ingestion backend on **Firebase and GCP**